

VIGNESHWAR KARUNANITHI

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ABOUT ME

An AI/ML Engineer with a master's degree in robotics and 3 years of experience in developing deep learning solutions for computer vision and embedded-AI systems. With a multidisciplinary background in robotics, software, and applied AI, I'm eager to learn continuously and contribute to innovative technologies that serve real-world societal needs.

TECHNICAL SKILLS

Programming & Software: Python, C, C++, Git, Linux, Docker, AWS

Tobotics & Simulation tools: ROS2, Gazebo, MATLAB/Simulink, FreeRTOS

Hardware platforms: Microcontrollers (ESP8266, ESP32, STM32, MSP430, Arduino Uno), Raspberry Pi, Jetson development boards (Nano, TX2, AGX Orin), XBee modules.

Computer Vision & AI: PyTorch, TensorFlow, OpenCV, Scikit-Learn, Detectron2, Visual-Inertial SLAM, Vision Transformers (ViT), GANs, Numpy, Pandas, Weights & Biases

PROFESSIONAL EXPERIENCE

FUGRO N.V.

Den Haag, NL

Graduate Research Student - Master's Thesis

JANUARY 2025 - PRESENT

The master's thesis is a collaboration between the University of Twente and Fugro N.V., focusing on domain adaptive object detection for maritime simulation images to leverage simulation data for real-world maritime object detection.

- Researched domain adaptation techniques to improve object detection performance when transferring simulated to real-world maritime images with differing visual domains.
- Developed simulation pipelines and annotation workflows to generate a synthetic maritime dataset covering varied environmental conditions and object categories.
- Developed an end-to-end pipeline combining adversarial learning, teacher-student training, and explainability analysis for robust cross-domain model development using Detectron2. Evaluated using COCO metrics.

FUGRO N.V.

Den Haag, NL

Computer Vision Intern

SEPTEMBER 2024 - DECEMBER 2024

- Worked on the object detection module of an Unmanned Surface Vessel (USV).
- Conducted feasibility analysis on the utilisation of simulation engines for the autonomous navigation task.
- Conducted research and performed a comparative study on Image-to-Image translation models to bridge the gap between simulation and real image domains.
- Evaluated image-to-image translation techniques, including GAN-based, contrastive learning, and stable diffusion models, to enhance the realism of simulation images.
- Developed a ROS2-based communication bridge between a maritime simulator and the navigation stack of the USV.

L&T TECHNOLOGY SERVICES

Mysore, India

Embedded AI Engineer

MAY 2021 - JULY 2023

Worked as part of the embedded-AI team of the Industrial Products business unit. Worked on multiple client projects and proof-of-concept projects in the domains of artificial intelligence, computer vision and embedded systems.

Vision-based fire and smoke detection:

- Developed a method and a system for detecting and estimating the propagation path of fire and smoke.
- Developed an image processing pipeline with colour segmentation and motion detection capabilities.
- Implemented a Kalman filter-based algorithm to estimate and track the propagation path of fire and smoke.
- Designed and deployed an end-to-end system for real-time inference on the Jetson Nano edge device.
- Optimised system performance for low-latency, high-reliability applications in safety-critical environments using TensorRT.

Benchmarking AI models:

- Analysed and benchmarked object detection models on standard computers and resource-constrained microcontrollers (Renesas RA6M5).
- Converted TensorFlow models to TensorFlow Lite (TFLite) and further optimised them by generating flat buffer arrays for microcontroller deployment.

- Implemented quantisation to reduce model footprint while preserving inference accuracy on edge devices.

Motor health monitoring:

- Researched fault types in BLDC motors and executed experiments to simulate failure scenarios.
- Collected experimental time-series vibration data using BMA400 accelerometer and applied Fast Fourier Transform (FFT) based signal analysis for feature extraction.
- Designed and trained machine learning models to detect faults in BLDC motors using time-series accelerometer data.
- Evaluated model performance and deployed the solution on a Raspberry Pi for real-time edge-based monitoring.

EDUCATION

UNIVERSITY OF TWENTE

Enschede, NL

Master of Science in Robotics with specialisation in Algorithms and Software AI

SEPTEMBER 2023 - PRESENT

- GPA - 8.02
- Relevant coursework - Image Processing and Computer Vision, Natural Language Processing, Deep Learning, Optimal Control, Robot perception, cognition and navigation, Advanced Software development for robotics, Data Science, Optimal Estimation of Dynamic Systems, Advanced Computer Vision and Pattern Recognition, AI for Autonomous Robots, Deep Learning for 3D Medical Imaging.

NATIONAL INSTITUTE OF TECHNOLOGY TIRUCHIRAPPALLI

Tamil Nadu, IN

Bachelor of Technology in Instrumentation and Control

AUGUST 2017 - MAY 2021

- CGPA - 8.76
- Minor in Computer Science. Relevant course - Digital systems, Microcontrollers and microprocessors, Medical instrumentation, sensors and transducers, micro-electro mechanical systems.

PROJECTS AND RESEARCH

Enhancing the robustness of object detection and tracking through multimodal integration

University of Twente

2024

- Developed a real-time multimodal object detection and tracking pipeline by integrating monocular camera images with 2D LiDAR slices, using YOLOv8 and DeepSORT for detection and tracking.
- Designed a lightweight depth estimation module by projecting 3d Lidar data into optimal 2D slices and associating them with image coordinates, significantly reducing computational overhead.
- Enhanced detection robustness in dynamic and adverse environments by evaluating on the NuScenes dataset with diverse weather and occlusion scenarios, achieving reliable tracking and distance estimation.

Automated Cardiac Diagnosis Challenge - Study of U-Net and MERIT for Cardiac MRI Segmentation

University of Twente

2024

- Conducted a comparative performance analysis of CNN-based U-Net and Transformer-based MERIT models for medical image segmentation on the ACDC dataset.
- Implemented robust preprocessing and data augmentation pipelines using MONAI and trained both models with optimised hyperparameters for 2D segmentation tasks.

IOT-based remote water quality monitoring system

National Institute of Technology, Tiruchirappalli

2021

- Designed and built a real-time water quality monitoring system using Arduino UNO, ESP8266 and XBee modules, integrating TDS, pH and temperature sensors to monitor the quality and hardness of water.
- Enabled wireless data transmission using RF communication with the MQTT protocol for efficient, low-latency data transfer.
- Designed a web-based dashboard for remote monitoring and visualisation of sensor data trends.

LANGUAGES

ENGLISH (Professional Fluency) | DUTCH (A1)

AWARDS AND HONOURS

- Filed a patent - Vision-based fire and smoke detection and propagation path estimation (IN202341018213).
- Winner - Tech Expression (annual technology innovation competition within L&T Technology Services).
- Employee of the Month - L&T Technology Services.